# Tripartite counterfactual entanglement distribution

**Yuanyuan Chen,[1] Xuemei Gu,[1] Dong Jiang,[1] Ling Xie,[1] and Lijun Chen,[1*]**

[1] *State Key Laboratory for Novel Software Technology, Nanjing University, Nanjing, 210046, China*

[*] *chenlj@nju.edu.cn*

**Abstract:** We propose two counterfactual schemes for tripartite entanglement distribution without any physical particles travelling through the quantum channel. One scheme arranges three participators to connect with the absorption object by using switch. Using the "chained" quantum Zeno effect, three participators can accomplish the task of entanglement distribution with unique counterfactual interference probability. Another scheme uses Michelson-type interferometer to swap two entanglement pairs such that the photons of three participators are entangled. Moreover, the distance of entanglement distribution is doubled as two distant absorption objects are used. We also discuss the implementation issues to show that the proposed schemes can be realized with current technology.



**OCIS codes:** (270.5565) Quantum communications; (270.5585) Quantum information and processing.

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## References and links

1. W. K. Wootters, "Entanglement of formation of an arbitrary state of two qubits," Phys. Rev. Lett. **80**, 2245 (1998).
2. R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, "Quantum entanglement," Rev. Mod. Phys. **81**, 865 (2009).
3. D. Bouwmeester, J.-W. Pan, K. Mattle, M. Eibl, H. Weinfurter, and A. Zeilinger, "Experimental quantum teleportation," Nature **390**, 575–579 (1997).
4. M. Riebe, H. Häffner, C. Roos, W. Hänsel, J. Benhelm, G. Lancaster, T. Körber, C. Becher, F. Schmidt-Kaler, D. James, and R. Blatt, "Deterministic quantum teleportation with atoms," Nature **429**, 734–737 (2004).
5. G. Vidal and R. F. Werner, "Computable measure of entanglement," Phys. Rev. A **65**, 032314 (2002).
6. T. Jennewein, C. Simon, G. Weihs, H. Weinfurter, and A. Zeilinger, "Quantum cryptography with entangled photons," Phys. Rev. Lett. **84**, 4729 (2000).
7. W. Tittel, J. Brendel, H. Zbinden, and N. Gisin, "Quantum cryptography using entangled photons in energy-time bell states," Phys. Rev. Lett. **84**, 4737 (2000).
8. A. M. Lance, T. Symul, W. P. Bowen, B. C. Sanders, and P. K. Lam, "Tripartite quantum state sharing," Phys. Rev. Lett. **92**, 177903 (2004).
9. J. Jing, J. Zhang, Y. Yan, F. Zhao, C. Xie, and K. Peng, "Experimental demonstration of tripartite entanglement and controlled dense coding for continuous variables," Phys. Rev. Lett. **90**, 167903 (2003).
10. N. Kiesel, C. Schmid, G. Tóth, E. Solano, and H. Weinfurter, "Experimental observation of four-photon entangled dicke state with high fidelity," Phys. Rev. Lett. **98**, 063604 (2007).
11. T. S. Cubitt, F. Verstraete, W. Dür, and J. I. Cirac, "Separable states can be used to distribute entanglement," Phys. Rev. Lett. **91**, 037902 (2003).
12. S. Perseguers, J. I. Cirac, A. Acín, M. Lewenstein, and J. Wehr, "Entanglement distribution in pure-state quantum networks," Phys. Rev. A **77**, 022308 (2008).
13. J.-W. Pan, D. Bouwmeester, H. Weinfurter, and A. Zeilinger, "Experimental entanglement swapping: Entangling photons that never interacted," Phys. Rev. Lett. **80**, 3891 (1998).

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14. G. Fauconnier, "Analogical counterfactuals," In G. Fauconnier and E. Sweetser (Eds.), Spaces, Worlds, and Grammar, The University of Chicago Press, pp. 57–90 (1996).
15. A. C. Elitzur and L. Vaidman, "Quantum mechanical interaction-free measurements," Found. Phys. **23**, 987–997 (1993).
16. O. Hosten, M. T. Rakher, J. T. Barreiro, N. A. Peters, and P. G. Kwiat, "Counterfactual quantum computation through quantum interrogation," Nature **439**, 949–952 (2006).
17. T.-G. Noh, "Counterfactual quantum cryptography," Phys. Rev. Lett. **103**, 230501 (2009).
18. H. Salih, Z.-H. Li, M. Al-Amri, and M. S. Zubairy, "Protocol for direct counterfactual quantum communication," Phys. Rev. Lett. **110**, 170502 (2013).
19. Q. Guo, L.-Y. Cheng, L. Chen, H.-F. Wang, and S. Zhang, "Counterfactual entanglement distribution without transmitting any particles," Opt. Express **22**, 8970–8984 (2014).
20. L. Vaidman, "Impossibility of the counterfactual computation for all possible outcomes," Phys. Rev. Lett. **98**, 160403 (2007).
21. L. Vaidman, "Comment on "Protocol for direct counterfactual quantum communication"," Phys. Rev. Lett. **112**, 208901 (2014).
22. H. Salih, Z.-H. Li, M. Al-Amri, and M. S. Zubairy, "Salih et al. Reply," Phys. Rev. Lett. **112**, 208902 (2014).
23. L.-M. Duan, M. Lukin, J. I. Cirac, and P. Zoller, "Long-distance quantum communication with atomic ensembles and linear optics," Nature **414**, 413–418 (2001).
24. L. C. Venuti, C. D. E. Boschi, and M. Roncaglia, "Long-distance entanglement in spin systems," Phys. Rev. Lett. **96**, 247206 (2006).
25. T. Honjo, S. W. Nam, H. Takesue, Q. Zhang, H. Kamada, Y. Nishida, O. Tadanaga, M. Asobe, B. Baek, R. Hadfield, S. Miki, M. Sasaki, Z. Wang, K. Inoue and Y. Yamamoto, "Long-distance entanglement-based quantum key distribution over optical fiber," Opt. Express **16**, 19118–19126 (2008).
26. H. A. Shenoy, R. Srikanth, and T. Srinivas, "Semi-counterfactual cryptography," EPL-Europhys. Lett. **103**, 60008 (2013).
27. C. Hu, W. Munro, J. OBrien, and J. Rarity, "Proposed entanglement beam splitter using a quantum-dot spin in a double-sided optical microcavity," Phys. Rev. B **80**, 205326 (2009).
28. C. Bonato, F. Haupt, S. S. Oemrawsingh, J. Gudat, D. Ding, M. P. van Exter, and D. Bouwmeester, "Cnot and bell-state analysis in the weak-coupling cavity qed regime," Phys. Rev. Lett. **104**, 160503 (2010).
29. Y. Cao, Y.-H. Li, Z. Cao, J. Yin, Y. Chen, X. Ma, C.-Z. Peng, and J.-W. Pan, "Direct counterfactual communication with single photons," in "CLEO: QELS_Fundamental Science," (Optical Society of America, 2014), pp. FM4A–6.

## 1. Introduction

Entanglement [1, 2] between two distant locations is an essential resource for most remarkable quantum processing tasks, such as quantum teleportation [3, 4], quantum computation [5], quantum cryptography [6, 7] and so on. Entanglement distribution is a key procedure to realize the remote establishment of entanglement [8–10]. In the past decades, entanglement distribution has been researched both theoretically and experimentally. Cubitt *et al.* [11] showed that two distant particles can be entangled by sending a third particle that is never entangled with the other two. Similarly, two particles can become entangled by continuous interaction with a highly mixed mediating particle. Perseguers *et al.* [12] showed any two stations of a network can share an entanglement pair if the effective probability for the quantum errors is below a certain threshold, which is achieved by a local encoding of the qubits and a global bit-flip correction. Pan *et al.* [13] experimentally entangled freely propagating particles that never physically interacted with one another or which have never been dynamically coupled by any other means.

Recently, a striking phenomenon that results from quantum mechanics is counterfactual quantum communication [14]. It can accomplish the communication task without any particles travelling through the quantum channel. The counterfactual quantum communication originates from the interaction-free measurement proposed by Elitzur and Vaidman [15]. Hosten *et al.* [16] demonstrated counterfactual quantum computation and used a novel "chained" quantum Zeno effect (CQZE) to boost the counterfactual interference probability to unity. Subsequently, Noh [17] proposed a counterfactual quantum cryptography scheme to distribute a secret key without transmitting the signal through the quantum channel. Salih *et al.* [18] presented a direct

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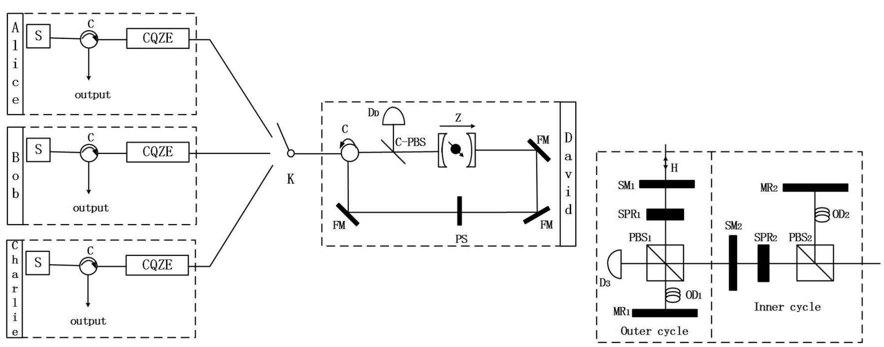
counterfactual quantum communication protocol to transmit a classical bit directly by using the CQZE system. Recently, Guo *et al.* [19] proposed a counterfactual protocol for entanglement distribution with no physical particles travelling between the two remote participants. However, Vaidman [20,21] argued that the possible outcomes of CQZE are not all counterfactual. In case when there is no blockade, the possibility for finding the photon by a nondemolition measurement on the transmission channel is one. While Salih [22] claimed that any weak measurement would affect the interference in the inner interferometer, which directly in conflict with the predictions of quantum mechanics. Up to now, the term "counterfactual" is still a subject of heated debate.

In this work, we propose two counterfactual schemes for tripartite entanglement distribution. Inspired by the basic idea of bilateral counterfactual entanglement distribution [19], we arrange three participators to connect with David (the absorption object) by using switch. Thus the photons of three participators are all entangled with the absorption object. Once David broadcasts his measurement result about the absorption object, the three participators know which entanglement is distributed. Another scheme uses Michelson-type interferometer to swap two entanglement pairs such that the photons of three participators are entangled. Moreover, as two remote absorption objects are used in this scheme, the distance of entanglement distribution is doubled.

## 2. Tripartite counterfactual entanglement distribution

In this section, we present a tripartite counterfactual counterfactual entanglement distribution scheme by using switch. To evaluate this scheme, the performance is discussed based on numerical analysis.

### 2.1. Protocol for tripartite counterfactual entanglement distribution




(a) (b)

Fig. 1. Experimental setup of tripartite counterfactual entanglement distribution. (a) Three participators connect with David (the absorption object) by using switch $K$. $S$ stands for single-photon source, $C$: optical circulator, $D_D$: photon detector, *C-PBS*: polarizing beam splitter in the circular basis, $FM$: Faraday mirror, *PS* is the phase shifter used to perform the transformation $|\phi\rangle \rightarrow -|\phi\rangle$, CQZE represents the "chained" quantum Zeno effect. (b) Experimental setup of "chained" quantum Zeno effect. *SM*: switchable mirror, *SPR*: switchable polarization rotator, $D_3$: photon detector, *OD*: optical delay.

As shown in Fig. 1, Alice, Bob and Charlie connect with David by a nested Michelson interferometer, respectively. Similar to [19], David first chooses to connect with Alice by using switch $K$. Alice triggers the single-photon source and sends a photon to the interferometer. The

CQZE system works as: $|H\rangle \to |H\rangle$ when the absorption object is absent, while $|H\rangle \to |V\rangle$ when the absorption object is present. Moreover, in both cases, the photon does not travel through the quantum channel. After the CQZE system, the state reads as:

$$|\phi\rangle_m = \frac{X_m|0\rangle_a|0\rangle_0 + Y_m|0\rangle_a|1\rangle_o + Z_m|1\rangle_a|1\rangle_o}{\sqrt{2}} \tag{1}$$

where $|0\rangle_a$ and $|1\rangle_a$ represent different polarization states, $|0\rangle_o$ and $|1\rangle_o$ represent the absence and presence of the absorption object, respectively, the parameters $X_m$, $Y_m$ and $Z_m$ denote the probability amplitudes and satisfy the recursion relations [19]:

$$\begin{aligned} X_m &= X_{m-1}\cos\theta \\ Y_m &= Y_{m-1}\cos\theta - Z_{m-1} sin\theta \\ Z_m &= (Y_{m-1}\sin\theta - Z_{m-1} cos\theta) cos^n\vartheta \end{aligned} \tag{2}$$

where $\theta = \pi/2m$, $\vartheta = \pi/2n$, $X_1 = Y_1 = \cos\theta$ and $Z_1 = \sin\theta\cos^n\vartheta$, $m$ and $n$ are the numbers of outer and inner cycles in the CQZE system. Next, we begin the second step, i.e., David chooses to connect with Bob. After the same procedure, the system state is:

$$\begin{aligned} |\phi\rangle_m =& \frac{1}{\sqrt{2}}(X_m^2|0\rangle_a|0\rangle_b|0\rangle_o + Y_m^2|0\rangle_a|0\rangle_b|1\rangle_o + Y_mZ_m|1\rangle_a|0\rangle_b|1\rangle_o + Y_mZ_m|0\rangle_a|1\rangle_b|1\rangle_o \\ &+ Z_m^2|1\rangle_a|1\rangle_b|1\rangle_o) \end{aligned} \tag{3}$$

The last step is that David chooses to connect with Charlie. He performs the same operation and the state of the system will become:

$$\begin{aligned} |\phi\rangle_m =& \frac{1}{\sqrt{2}}(X_m^3|0\rangle_a|0\rangle_b|0\rangle_c|0\rangle_o + Y_m^3|0\rangle_a|0\rangle_b|0\rangle_c|1\rangle_o + Y_m^2Z_m|1\rangle_a|0\rangle_b|0\rangle_c|1\rangle_o \\ &+ Y_m^2Z_m|0\rangle_a|1\rangle_b|0\rangle_c|1\rangle_o + Y_m^2Z_m|0\rangle_a|0\rangle_b|1\rangle_c|1\rangle_o + Y_mZ_m^2|1\rangle_a|1\rangle_b|0\rangle_c|1\rangle_o \\ &+ Y_mZ_m^2|1\rangle_a|0\rangle_b|1\rangle_c|1\rangle_o + Y_mZ_m^2|0\rangle_a|1\rangle_b|1\rangle_c|1\rangle_o + Z_m^3|1\rangle_a|1\rangle_b|1\rangle_c|1\rangle_o) \end{aligned} \tag{4}$$

In order to obtain perfect entanglement, we should choose suitable $m$ and $n$ to tune the probability amplitudes. Figure 2 shows the variation trend of the coefficients versus different values of $m$ and $n$. We can see that for large $m$ and $n$, $X_m^3 \to 1$, $Y_m^3 \to 0$, $Y_m^2Z_m \to 0$, $Y_mZ_m^2 \to 0$ and $Z_m^3 \to 1$. Here we choose the parameters to obtain the approximate fidelity got in [19]. Thus, $X_m^3 = 0.9597$, $Y_m^3 = 0$, $Y_m^2Z_m = 0$, $Y_mZ_m^2 = 0.0013$ and $Z_m^3 = 0.9418$ for $m = 100$ and $n = 4000$. The state in Eq. (4) can be approximately expressed as $|\phi\rangle_m \simeq \frac{|0\rangle_a|0\rangle_b|0\rangle_c|0\rangle_o + |1\rangle_a|1\rangle_b|1\rangle_c|1\rangle_o}{\sqrt{2}}$. After that, David performs a Hadamard transformation $\{|0\rangle \to \frac{|0\rangle+|1\rangle}{\sqrt{2}}, |1\rangle \to \frac{|0\rangle-|1\rangle}{\sqrt{2}}\}$ on the absorption object, the system state will become:

$$|\phi\rangle_m \to \frac{1}{\sqrt{2}}[(|0\rangle_a|0\rangle_b|0\rangle_c + |1\rangle_a|1\rangle_b|1\rangle_c)|0\rangle_o + (|0\rangle_a|0\rangle_b|0\rangle_c - |1\rangle_a|1\rangle_b|1\rangle_c)|1\rangle_o] \tag{5}$$

Then David measures the absorption object and broadcasts the measurement result. According to David′s announcement, Alice, Bob and Charlie can know which entanglement is distributed.

### 2.2. *Discussion for tripartite counterfactual entanglement distribution*

However, we note that $m$ and $n$ are much larger than those used in [19]. Suppose the electron-spin coherence time is $T^e$ and the distance between Alice (Bob, Charlie) and David is $L$. The

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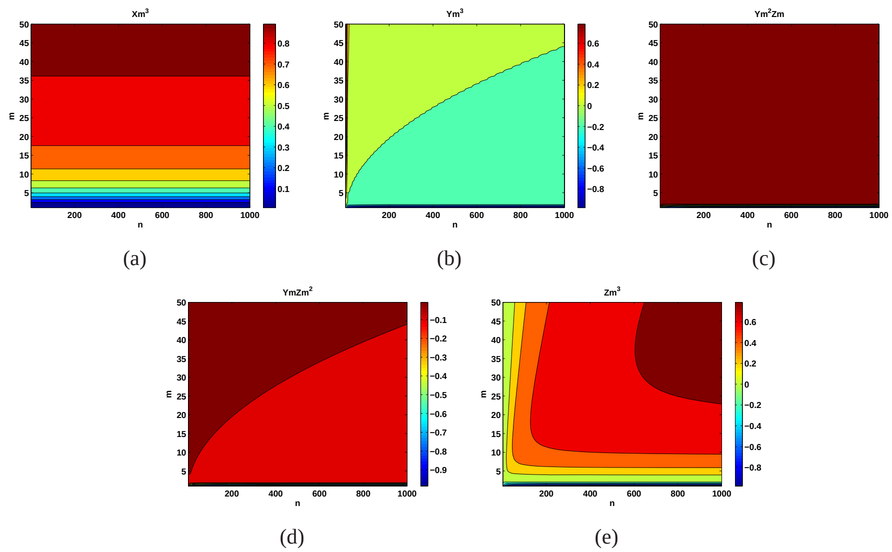


Fig. 2. The parameters $X_m^3$, $Y_m^3$, $Y_m^2 Z_m$, $Y_m Z_m^2$ and $Z_m^3$ in Eq.(4) versus the different values of $m$ and $n$. (a) $X_m^3$ approaches one with the increase of $m$ and does not change with $n$, (b) $Y_m^3$ is close to zero, (c) $Y_m^2 Z_m$ is also close to zero, (d) $Y_m Z_m^2$ is also close to zero, (e) $Z_m^3$ approaches one with the increase of $m$ and $n$

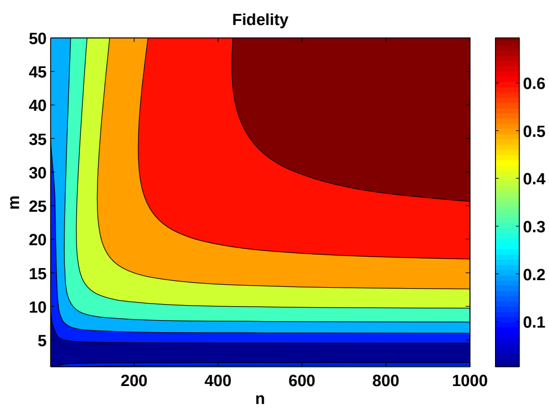


Fig. 3. The fidelity of tripartite counterfactual entanglement distribution versus different values of outer cycles $m$ and inner cycles $n$.

required time to accomplish the scheme is $t \approx 3nmL/c$ ($c$ is the speed of light), which should satisfy $T^e \geq 3nmL/c$. For given $T^e$, the distance of entanglement distribution is reduced as the values of $m$ and $n$ increase.

The perfect entanglement requires that the state of photon-electron system becomes a maximal GHZ-type entangled state, i.e., $|\psi'\rangle_m = \frac{1}{\sqrt{2}}(|0\rangle_a|0\rangle_b|0\rangle_c + |1\rangle_a|1\rangle_b|1\rangle_c)$. Hence, the fidelity of entanglement in the scheme can be obtained as $F = |\langle\psi'|\psi\rangle|^2 = \frac{1}{4}(X_m^3 + Z_m^3)^2$. We plot the fidelity change with $m$ and $n$, as shown in Fig. 3. For large $m$ and $n$, the fidelity can approach one. But if the values of $m$ and $n$ increase, photon loss may also increase, which is caused by linear optical elements. Fortunately, the photon loss does not affect the error key rate, but affects the efficiency of the scheme. Suppose that the photon loss rate in every outer cycle is $\varepsilon$, then the efficiency of the scheme will be $(1-\varepsilon)^{3m}$. Obviously, the efficiency will be reduced exponentially with the value of outer cycles increasing linearly.

The imperfections of the system will affect the efficiency. As discussed above, each BS used in CQZE has large reflectivity, e.g., $R_m = \cos^2\theta$ and $R_n = \cos^2\vartheta$. However, in practical applications, a slight error $s$ may exist, which results in operating the photon with an additional angle $\Delta\theta = s\theta$ and $\Delta\vartheta = s\vartheta$. In this case, we can re-evaluate the performance by substituting $\theta$ ($\vartheta$) with $\theta+\Delta\theta$ ($\vartheta+\Delta\vartheta$). As shown in Fig. 4, we plot the fidelity versus different values of $s$.

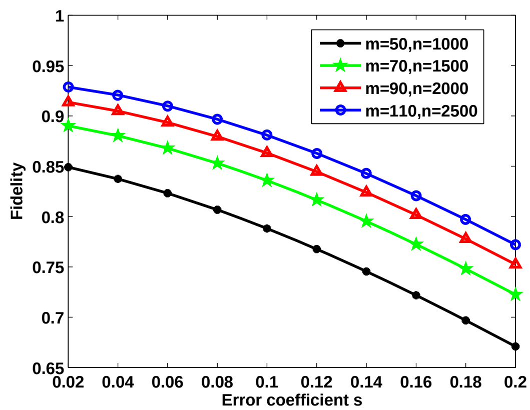


Fig. 4. The fidelity of tripartite counterfactual entanglement distribution versus the error coefficient $s$.

## 3. Long-distance tripartite counterfactual entanglement distribution

In order to prolong the distance of entanglement distribution [23–25], we arrange Charlie to use Michelson-type interferometer to swap two entanglement pairs.

### 3.1. *Protocol for long-distance tripartite counterfactual entanglement distribution*

As shown in Fig. 5, Alice, Bob and Charlie want to share one entanglement, then our scheme runs as follows.

1. Charlie triggers the single-photon source $S$, which emits a short optical pulse containing a single photon. The polarization state of the single-photon pulse is initially set to be $|H\rangle$. Then the single-photon is sent towards the beam splitter $BS_C$, which puts the photon in an equal superposition of being on path $a$ (leading to Alice) and path $b$ (leading to Bob). Here the transmissivity and reflectivity of the $BS_C$ are 50:50. After the $BS_C$, the state of the photon is:

$$|\psi\rangle = \frac{|0\rangle_a|1\rangle_b + |1\rangle_a|0\rangle_b}{\sqrt{2}} \tag{6}$$

where $|0\rangle$ and $|1\rangle$ represent the absence and presence of the photon in each path, respectively.

2. The left hand side of our scheme (Alice and path $a$ of Charlie) entangles with the absorption object $o$ (David), while the right hand side of our scheme (Bob and path $b$ of Charlie) entangles

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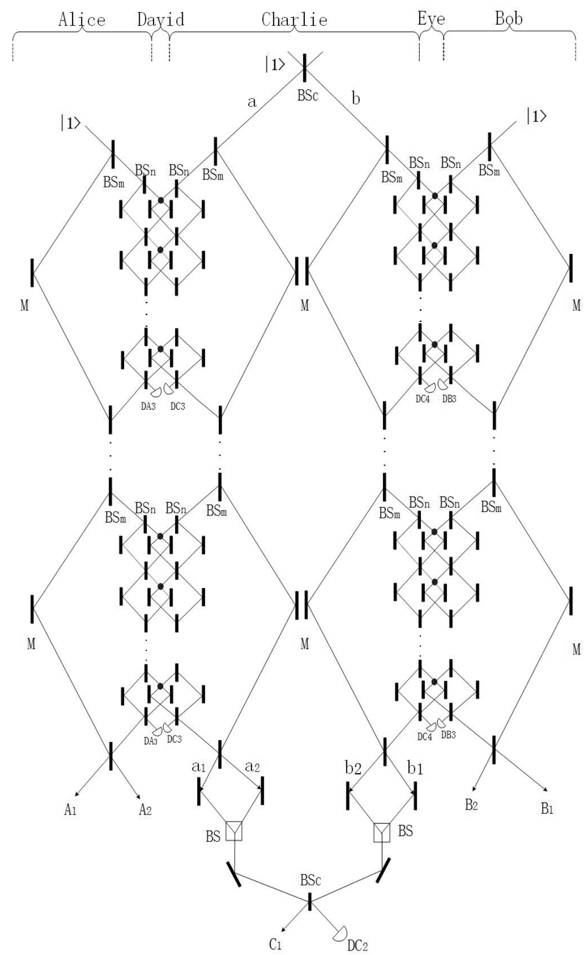


Fig. 5. Principle of long-distance tripartite counterfactual entanglement distribution. $BS_C$ is a 50:50 beam splitter, $BS_m$ is the unbalanced beam splitter with reflectivity $\cos^2\theta$ and transmissivity $\sin^2\theta$ ($\theta = \pi/2m$),while $BS_n$ is also the unbalanced beam splitter with reflectivity $\cos^2\vartheta$ and transmissivity $\sin^2\vartheta$ ($\vartheta = \pi/2n$), $DA_3$, $DC_2$, $DC_3$, $DC_4$ and $DB_3$ are photon detectors, $M$ is a normal mirror. One optical pulse is sent from Charlie and is then split by the $BS_C$. If the absorption objects are present, the superposition pass through the path $a_1$ and $b_1$, and the photons of Alice and Bob would go through path $A_2$ and $B_2$, respectively. In contrast, if the absorption objects are absent, the superposition pass through the path $a_2$ and $b_2$, and the photons of Alice and Bob would go through path $A_1$ and $B_1$, respectively. Then the superposition comes back to $BS_C$ (the Michelson-type interferometer).

with the absorption object $p$ (Eve). Then we get two entanglement pairs:

$$\begin{aligned}|\psi\rangle_{AC} &= \frac{1}{\sqrt{2}}[(|0\rangle_A|0\rangle_{C_A} + |1\rangle_A|1\rangle_{C_A})|0\rangle_o + (|0\rangle_A|0\rangle_{C_A} - |1\rangle_A|1\rangle_{C_A})|1\rangle_o] \\ |\psi\rangle_{BC} &= \frac{1}{\sqrt{2}}[(|0\rangle_B|0\rangle_{C_B} + |1\rangle_B|1\rangle_{C_B})|0\rangle_p + (|0\rangle_B|0\rangle_{C_B} - |1\rangle_B|1\rangle_{C_B})|1\rangle_p]\end{aligned} \tag{7}$$

3. The superpositions $|\varphi\rangle_{C_A}$ and $|\varphi\rangle_{C_B}$ from the entanglement pairs $|\psi\rangle_{AC}$ and $|\psi\rangle_{BC}$ come back to the $BS_C$ at Charlie's location. If $|\varphi\rangle_{C_A}$ is the same with $|\varphi\rangle_{C_B}$, the photon superposition

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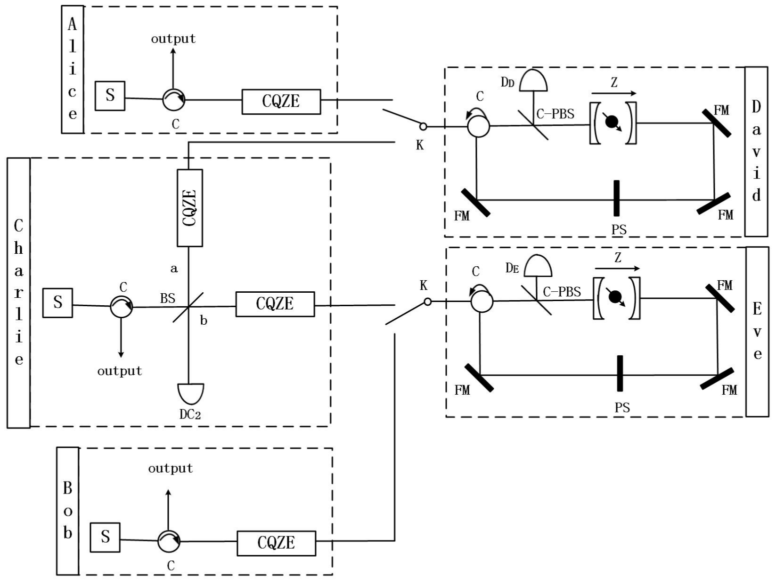


Fig. 6. Experimental setup of long-distance tripartite counterfactual entanglement distribution. $S$ stands for the single-photon source, $C$ is the optical circulator, $DC_2$, $D_D$ and $D_E$ are photon detectors, *C-PBS* is the polarizing beam splitter in the circular basis, *PS* is the phase shifter used to perform the transformation $|\phi\rangle \to -|\phi\rangle$, $FM$ is the Faraday mirror, CQZE represents the "chained" quantum Zeno effect.

leaves the interferometer going toward detector $DC_2$. If, however, $|\varphi\rangle_{C_A}$ differs from $|\varphi\rangle_{C_B}$, the interference does not take place and the photon would go through path $C_1$ and detector $DC_2$ with equal probability [26]. Thus if $DC_2$ clicks, we consider the entanglement distribution fails. Otherwise, the system state after the $BS_C$ is ($|0\rangle_{C_A}|1\rangle_{C_B} \to |0\rangle_C, |1\rangle_{C_A}|0\rangle_{C_B} \to |1\rangle_C$):

$$\begin{aligned}
|\psi\rangle =& \frac{1}{2}[(|0\rangle_A|1\rangle_B|0\rangle_C + |1\rangle_A|0\rangle_B|1\rangle_C)|0\rangle_o|0\rangle_p + (|0\rangle_A|1\rangle_B|0\rangle_C - |1\rangle_A|0\rangle_B|1\rangle_C)|0\rangle_o|1\rangle_p \\
& + (|0\rangle_A|1\rangle_B|0\rangle_C - |1\rangle_A|0\rangle_B|1\rangle_C)|1\rangle_o|0\rangle_p + (|0\rangle_A|1\rangle_B|0\rangle_C + |1\rangle_A|0\rangle_B|1\rangle_C)|1\rangle_o|1\rangle_p] \\
=& \frac{1}{2}[(|0\rangle_A|1\rangle_B|0\rangle_C + |1\rangle_A|0\rangle_B|1\rangle_C)(|0\rangle_o|0\rangle_p + |1\rangle_o|1\rangle_p) + (|0\rangle_A|1\rangle_B|0\rangle_C - |1\rangle_A|0\rangle_B|1\rangle_C) \\
& (|0\rangle_o|1\rangle_p + |1\rangle_o|0\rangle_p)]
\end{aligned} \tag{8}$$

4. David and Eve measure their absorption objects and broadcast the measurement results. If the measurement result is $|0\rangle_o|0\rangle_p$ or $|1\rangle_o|1\rangle_p$, the system state of Alice, Bob and Charlie collapses to $|0\rangle_A|1\rangle_B|0\rangle_C + |1\rangle_A|0\rangle_B|1\rangle_C$. Otherwise, if the measurement result is $|0\rangle_o|1\rangle_p$ or $|1\rangle_o|0\rangle_p$, the system state of Alice, Bob and Charlie collapses to $|0\rangle_A|1\rangle_B|0\rangle_C - |1\rangle_A|0\rangle_B|1\rangle_C$.

### *3.2. Discussion for long-distance tripartite counterfactual entanglement distribution*

Here we implement long-distance tripartite entanglement distribution among Alice, Bob and Charlie without transmitting any physical particles through the quantum channel. Then the distributed entanglement can be used for quantum communication, quantum cryptography and so on in large scale.

The experimental setup of long-distance tripartite counterfactual entanglement distribution is described in Fig. 6. Alice, Bob and Charlie trigger the photon source and emit a single photon, respectively. The single photon of Charlie is split by beam splitter to path $a$ and $b$. Here David and Eve use the photon-QD-microcavity system to control and manipulate the presence and absence of the absorption objects. Thus, similar to [19], two entanglement pairs are distributed. Then the superposition from the two entanglement pairs comes back to the BS at Charlie′s location. According to the interference theory, the photons of Alice, Bob and Charlie are entangled. Due to the CQZE system, there is not any particles travelling through the quantum channel in the whole process.

The two entanglement pairs used in the scheme are the same with [19], so the parameters can be set as $m = 30$ and $n = 1000$. Moreover, the distance of entanglement distribution between Alice and Bob is doubled with the help of Charlie. Here, the fidelity of entanglement in our scheme can be obtained as $F = |\langle\psi'|\psi\rangle|^2 = \frac{1}{4}(X_m^2 + Z_m^2)^2$. As discussed before, the photon goes through path $C_1$ with probability $\frac{1}{4}$. So the efficiency of this scheme is $E = \frac{1}{4}F^2 = \frac{1}{64}(X_m^2 + Z_m^2)^4$. We plot the efficiency change with $m$ and $n$, as shown in Fig. 7. For large values of $m$ and $n$, the efficiency of this scheme can approach $\frac{1}{4}$.

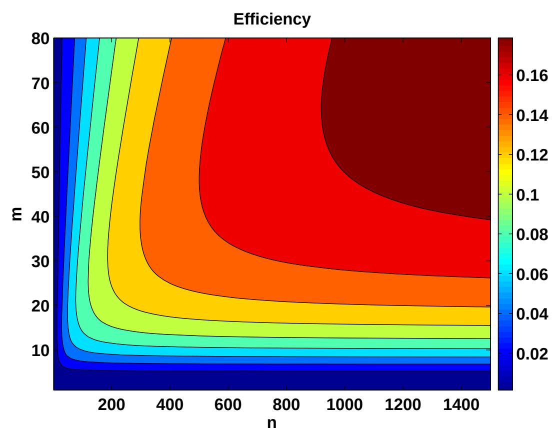


Fig. 7. Efficiency of long-distance tripartite counterfactual entanglement distribution versus different values of outer and inner cycles $m$ and $n$.

The noise in the transmission channel also affects the efficiency. Suppose the photon loss rate is $\varepsilon$, thus the efficiency will be $(\frac{1-\varepsilon}{4})^{2m}$. The efficiency is still reduced exponentially with the value of outer cycles increasing linearly. Compared with the above scheme, the efficiency of this scheme is lower.

Moreover, this scheme can be extended to longer distance by adding intermediate entanglement swapping. Suppose that the number of users is $N$, then the fidelity of distributed entanglement will become $F_N = (\frac{X_m^2+Z_m^2}{2})^N$. As shown in Fig. 8, we plot the fidelity change with the number of users $N$. When the number of users $N$ is increased, the fidelity of distributed entanglement is exponentially decreased.

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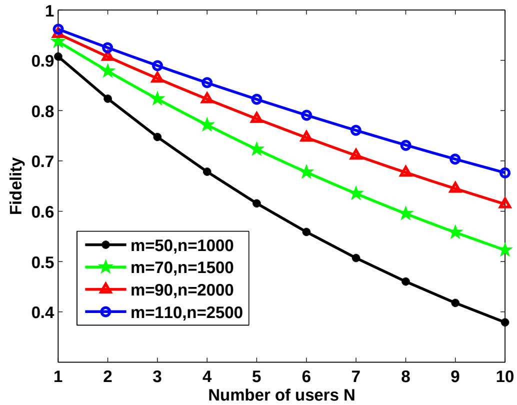


Fig. 8. The fidelity of long-distance tripartite counterfactual entanglement distribution versus different numbers of users.

As discussed above, a slight error $s$ may exist in the beam splitter. Therefore, the fidelity can be obtained by substituting $\theta$ ($\vartheta$) with $\theta + \Delta\theta$ ($\vartheta + \Delta\vartheta$). The final distributed entanglement fidelity of this scheme versus different values of $s$ is shown in Fig. 9. Compared with the above scheme, the fidelity of this scheme is higher in the case of the same error coefficient $s$.

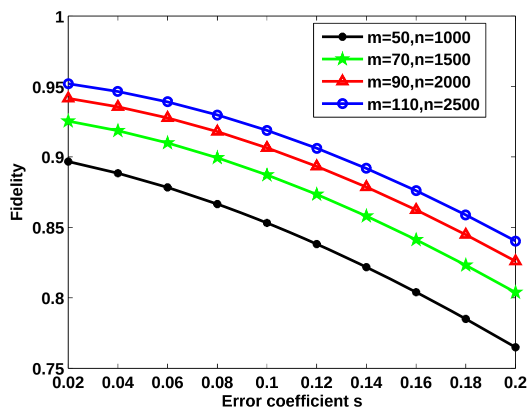


Fig. 9. The fidelity of long-distance tripartite counterfactual entanglement distribution versus the error coefficient $s$.

## 4. Implementation issues

In order to implement the proposed tripartite counterfactual entanglement distribution schemes, we need to realize two main operations: the quantum control and manipulate of the presence and absence of the absorption object and the realization of the two-cycle CQZE.

Recently, Hu *et al.* [27] proposed an entanglement beam splitter by using a quantum-dot spin in a double-side optical microcavity. The entanglement beam splitter can directly split a photon-spin product state into two constituent entangled states via transmission and reflection with high fidelity and high efficiency. Moreover, it has been recognized by Bonate *et al.* [28]. As discussed by Guo *et al.* [19], the photon-QD-microcavity system described in [27] is very

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suitable to serve as the quantum version of the absorption object.

Hosten *et al.* [16] used a novel “chained ” version of the quantum Zeno effect to implement counterfactual quantum computation with boosting the counterfactual interference probability to unity. Their strategy is to place a “Zeno scheme” inside another one, to avoid the photon transmitting over the quantum channel. This work maybe open the way for optical communication without photons transmitted physically. Using a single photon source, Cao *et al.* [29] experimentally successfully transfer a monochrome bitmap from one location to another by employing a nested version of the quantum Zeno effect.

In summary, the implementation issues used in our schemes can be realized in experiment. Thus our schemes are feasible using today′s technology, which makes commercial applications conceivable.

## 5. Conclusion

In conclusion, we propose two counterfactual schemes for tripartite entanglement distribution without any physical particles travelling through the quantum channel. Inspired by the bilateral counterfactual entanglement distribution, one scheme arranges three participators to connect with the absorption object by using switch. Thus, the three participators can accomplish the task of entanglement distribution with unique counterfactual interference probability by using the “chained” quantum Zeno effect. Another scheme uses Michelson-type interferometer to swap two entanglement pairs such that the distance of entanglement distribution is doubled. We also discuss the implementation issues to show that the proposed schemes can be realized with current technology.

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